Straw and plastic mulch effects on soil moisture retention in rain-fed sorghum: a two-season field trial

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Received 3 February 2027; accepted 18 May 2027

Abstract

Soil water shortage during grain filling is the main constraint on sorghum yield in semi-arid smallholder systems. We compared bare soil with straw mulch and black polyethylene film over two rainy seasons on a sandy loam. Both mulches raised volumetric water content in the top 30 cm by 4.6 to 6.3 percentage points and increased grain yield by 36 to 57 per cent. Straw mulch achieved most of the benefit of plastic film at a fraction of its cost and without residue disposal problems.

Keywords: mulching; soil water content; sorghum; semi-arid agriculture; water use efficiency

1. Introduction

Rain-fed sorghum is grown on more than two million hectares of semi-arid land in the region, mostly by farmers with less than two hectares. Rainfall is erratic, and dry spells of two to three weeks commonly coincide with flowering and grain filling. Under these conditions, much of the rain that does fall is lost by evaporation from the soil surface before the crop can use it.

Surface mulches reduce evaporation by shading the soil and slowing the exchange of water vapour with the air above it. Plastic film is effective but expensive and leaves fragments in the soil, whereas crop residues are cheap but compete with livestock feed. Few field studies in the region have compared the two under the same management.

This communication reports soil water content and grain yield from a two-season trial designed to answer a simple question: how much of the benefit of plastic film can a straw mulch deliver?

2. Materials and methods

2.1 Study site

The trial was conducted at the Maruni dryland research station at an elevation of 1,140 m. The soil is a sandy loam with 68% sand, 0.6% organic carbon and a pH of 6.4. Mean annual rainfall over the previous twenty years was 612 mm, falling mainly between November and April.

2.2 Experimental design

Three treatments were laid out in a randomised complete block design with four replicates. Plots measured 6 m by 5 m and were sown with an improved open-pollinated sorghum variety at a spacing of 75 cm by 20 cm. Straw mulch was applied at 5 t ha^{-1} two weeks after emergence; the polyethylene film was laid before sowing and the seed was planted through cut holes.

Volumetric soil water content was measured weekly at 0–30 cm with a calibrated capacitance probe, and grain was harvested from the central four rows of each plot. Data were analysed by analysis of variance, and means were separated by the least significant difference at the 5% level.

3. Results

Seasonal rainfall was 584 mm in 2025 and 471 mm in 2026, and the second season included a dry spell of 23 days at flowering. In both seasons, soil water content under the two mulches remained above that of bare soil from the third week after emergence until harvest.

Across both seasons, mulched plots held between 4.6 and 6.3 percentage points more water in the top 30 cm than bare plots. The difference between straw and film was smaller than the least significant difference in both seasons.

Grain yield followed the same pattern (Table 1). Relative to the control, straw mulch increased yield by 36% in 2025 and by 51% in the drier 2026 season. The corresponding increases under plastic film were 46% and 57%.

Table 1. Mean volumetric soil water content during grain filling and grain yield by treatment and season (mean ± standard deviation, n = 4).

Treatment	Soil water content, 0–30 cm (%)		Grain yield (t ha^{-1})	
	2025	2026	2025	2026
Bare soil (control)	14.2 ± 1.1	12.8 ± 1.3	1.62	1.35
Straw mulch, 5 t ha^{-1}	18.9 ± 0.9	17.4 ± 1.0	2.21	2.04
Black polyethylene film	20.3 ± 1.2	19.1 ± 1.4	2.36	2.12
LSD (p = 0.05)	*1.6*	*1.8*	*0.19*	*0.22*

4. Discussion

The larger relative gain in the drier season suggests that mulching is most valuable precisely when water is scarcest. Straw delivered between 80% and 90% of the yield gain achieved with film, while costing about one fifth as much per hectare when valued at local fodder prices. Film also left visible fragments in the soil after a single season.

The main limitation of straw is its alternative value as dry-season feed. Farmers who keep cattle may prefer to mulch only the part of the field most prone to drought, a practice that deserves its own trial.

5. Conclusions

Straw mulch at 5 t ha^{-1} is a low-cost way to conserve soil water and raise sorghum yields on sandy soils in semi-arid conditions. Its benefit should be weighed against the feed value of the straw on each farm.

Acknowledgements

We thank the field staff of the Maruni station for their care in maintaining the plots, and the farmers of Maruni village for their advice.

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